A Situation Aware Generative Agent Based on Psychological Characteristics of Situation

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***Abstract*—Social support agents need to understand not only tasks, but also the psychological characteristics of the situations to truly assist users in their daily tasks. Recent work has shown that modelling Psychological Characteristics of Situations (PCS) can improve decision-making in personal assistant agents. In this paper, I present SAGA, a Situation Aware Generative Agent that uses large language models (LLMs) to extract PCS from natural language descriptions and integrates this information into a classical machine learning pipeline for priority prediction. The agent is grounded in the DIAMONDS taxonomy and relies on a pre-trained Random Forest model to assess the priority of social situations. In addition, SAGA provides explanations for its decision and supports a simple negotiation mechanism when the user disagrees with the proposed decision. We evaluate several small and mid-sized LLMs on a dataset of 2,225 situations, using Mean Absolute Error and Mean Squared Error to assess PCS extraction and priority prediction. Overall, this work demonstrates that combining psychological situation models with generative language models enables flexible and explainable social decision-making in support agents and provides a practical baseline for future research on situation-aware personal assistants.**

***Index Terms*—Situation Awareness, Psychological Characteristics of Situations, DIAMONDS Taxonomy, Large Language Models, Social Support Agents, Explainable AI, Intelligent Agent System**

I. INTRODUCTION

Intelligent assistant systems are increasingly being used to help users manage their daily tasks, such as time management, meeting planning, or task organisation. However, these decisions are rarely based only on objective constraints such as meeting time or location. Choosing which appointment to prioritise often depends on social and psychological factors. A meeting with a close colleague, an important supervisor, or a stressed friend may be perceived very differently, even if they appear similar from a logistical point of view.

This is why recent research argues that intelligent agents should be able to understand situations in a way that is closer to human perception [1]. Rather than relying only on objective features, agents should consider how a situation is experienced and interpreted. Psychological Characteristics of Situations (PCS) are a useful model for representing and analysing social situations. In particular, the DIAMONDS framework describes situations according to several dimensions such as duty, mating potential, or sociality, which have been shown to influence human behaviour and decision-making. In the reference paper, PCS are used to support social situation comprehension in intelligent agents, showing that these representations can improve the prioritisation and interpretation of social events. However, the original approach relies on manually obtained situation descriptions.

At the same time, Large Language Models (LLMs) have recently shown strong capabilities in natural language understanding, social information extraction and semantic reasoning. These models can process textual descriptions and understand social or contextual assumptions that are difficult to model using traditional NLP techniques. This makes LLMs a promising tool for automatically extracting PCS from natural language descriptions of social situations.

In this project, I propose SAGA, which stands for Situation-Aware Generative Agent, an intelligent agent framework that integrates LLM-based PCS extraction into a decision-support and interaction pipeline. The agent analyses textual descriptions of social situations, automatically extracts psychological characteristics using LLMs and predicts the priority of appointments based on these characteristics.

Unlike the reference model, SAGA is designed as an interactive agent capable of explaining its decisions to the user and engaging in a negotiation process when its recommendations are rejected.

The contributions of this work are twofold. Firstly, I implement and evaluate the automatic extraction of PCSs from textual social descriptions using LLM and compare them with the ML models in the paper. Secondly, I integrate these components into an agent pipeline built using LangGraph, enabling the generation of explanations and adaptive negotiation with the user, all in the form of a conversational agent. This work does not aim to replace the theoretical model of the article but proposes an extension and a more concrete implementation.

II. Background and Existing Work

Understanding social situations is a central challenge for intelligent support agents, especially when these agents must assist users in daily decision-making. Classical agent systems often struggle to capture the subjective and contextual aspects of human situations. In contrast, humans interpret situations based on how they perceive social roles, expectations, obligations, or emotional tone. This gap between computational representations and human perception motivates the use of psychologically grounded models of situations.

One influential line of research addresses this issue through the concept of Psychological Characteristics of Situations (PCS). Rather than describing situations in terms of surface features, PCS try to capture how situations are experienced and interpreted by individuals. Early work by Edwards and Templeton [2] showed that people consistently perceive situations according to a limited set of psychological dimensions. Building on this idea, Rauthmann et al. proposed the DIAMONDS framework, which defines eight major dimensions of situation characteristics, including Duty, Intellect, Adversity, Mating, Positivity, Negativity, Deception, and Sociality [3].

- Duty : situations where a job must be done, minor details are important, and rational thinking is called for
- Intellect : situations that afford an opportunity to demonstrate intellectual capacity
- Adversity : situations where you or someone else are (potentially) being criticized, blamed, or under threat
- Mating : situations where potential romantic partners are present, and physical attractiveness is relevant
- pOsitivity : playful and enjoyable situations, which are simple and clear-cut
- Negativity : stressful, frustrating, and anxiety-inducing situations.
- Deception : situations where someone might be deceitful. These situations may cause feelings of hostility.
- Sociality : situations where social interaction is possible, and close personal relationships are present or have the potential to develop.

These dimensions have been empirically validated and shown to relate to behaviour, motivation, and decision-making. The reference paper by Kola [1] applies the DIAMONDS framework to the design of intelligent support agents. In this work, PCS are used as an intermediate representation that allows agents to reason about social situations in a structured and psychologically meaningful manner. The core idea is that by mapping situations to PCS vector, an agent can better estimate the importance or priority of different social events. This approach is related to the notion of situation awareness as defined by Endsley, where effective decision-making depends on perceiving, understanding, and projecting the state of the environment [4].

In the proposed model of the reference paper, each social situation is represented as a vector of PCS values. These values are then used as input features for a predictive model that estimates meeting priority. A Random Forest Regressor is employed for this task, due to its ability to model non-linear relationships between PCS dimensions.

The experimental results reported by Kola show that PCS-based representations enable meaningful priority predictions and outperform simpler models that rely only on situational characteristics [1]. The researchers benchmarked numerous models on each of the PCSs, and the model that achieved the best results was the Random Forest Regressor. I therefore also used this model for my agent, particularly for the prediction module.

In the experimental setup of the reference paper, PCS are derived from structured data and annotated situation descriptions, which allows for credible evaluation and comparison. I used the same dataset compiled by the researchers during their studies for my benchmark. However, in everyday interactions, users do not express their situations in terms of predefined psychological dimensions. Instead, they describe them using free text, often informal, incomplete or subjective. This creates a mismatch between the structured PCS representation used by the model and the natural language input produced by users.

In addition, the reference model focuses on prediction and does not explicitly address user interaction beyond presenting a priority outcome. Previous work on social situation awareness for behaviour support agents has emphasized the importance of interaction and adaptation in such systems [5], [6].

Despite these limitations, the reference paper provides a strong basis for further work. It demonstrates that PCS constitute an effective intermediate representation for social situation comprehension and priority prediction, and it provides publicly available data and evaluation results that can be reused as a baseline [1]. In this project, the PCS-based model and its Random Forest metrics are adopted to ensure comparability, while extending the original approach toward automatic PCS extraction from text and interactive agent behaviour.

III. The SAGA Architecture

This section presents the proposed architecture, called SAGA. The objective of SAGA is to extend the reference model by enabling automatic situation analysis from text and by integrating this analysis into an interactive agent pipeline. The agent is designed to support users in prioritising social situations, such as meetings, while remaining transparent and open to user feedback. This approach separates clearly the roles between components. Large Language Models are used to interpret natural language descriptions and extract psychological characteristics of situations. Classical machine learning models are then used to predict priority based on these characteristics. Finally, an agent architecture coordinates perception, decision-making, explanation, and negotiation.

A. Data and Situation Descriptions

The experiments rely on the dataset provided with the reference paper [1], which contains annotated social situations represented using Psychological Characteristics of Situations (PCS) and associated priority values. These PCS values serve as ground truth and are defined on a Likert scale from 1 to 7 for each DIAMONDS dimension. To enable textual analysis of the situation, since during a conversation the user gives a brief description of the situation rather than its characteristics, textual descriptions were generated from the structured data. These descriptions were produced using Gemini, a large and powerful language model, based on the social features available in the CSV file and the associated PCS values. As a result, the descriptions are synthetic but grounded in real annotated situations. Gemini used as inspiration the examples of situations given in the survey attached to the dataset.

B. Extraction of Psychological Characteristics of Situations using Large Language Models

The first core component of SAGA is the automatic extraction of PCS from textual descriptions. This task is treated as a perception problem. Given a free-text description of a situation, the agent estimates the values of the eight DIAMONDS dimensions: Duty, Intellect, Adversity, Mating, Positivity, Negativity, Deception, and Sociality. This extraction is performed using Large Language Models with a constrained prompt. The prompt explicitly defines each DIAMONDS dimension and requires the model to output a valid JSON object containing integer values between 1 and 7. No explanations or additional text are allowed in the output. This strict format reduces ambiguity. Several models were evaluated for this task, including models of different sizes and architectures, mostly locally. The selected models include Gemma 2 (2B and 9B), Mistral 7B, Phi-3 (3.8B) and LLaMA 3.2 (3B). All models were tested using the same prompt and parameters, without adjustment, to allow for a fair comparison.

C. Priority Prediction Using Psychological Characteristics of Situations

Once PCS values are obtained, the priority of a situation is predicted using the Random Forest Regressor. This model is the same as the one developed in the reference paper and attached to the dataset. The PCS values extracted by the LLM are used as input features, and the model outputs a continuous priority score between 1 and 7. This design choice reflects a clear separation of responsibilities. The LLM is used as a perception module that translates unstructured text into a structured psychological representation. The final decision is made by a classical machine learning model that is easier to interpret and already validated in previous work. This avoids relying on the LLM as a black-box decision-maker.

D. Large Language Model Benchmark

To evaluate the quality of PCS extraction and its impact on priority prediction, a benchmarking pipeline was implemented. For each situation in the dataset, PCS values were extracted using a given LLM, and the predicted PCS were compared to the ground truth annotations using Mean Absolute Error (MAE) and Mean Squared Error (MSE). The extracted PCS were then passed to the Random Forest model to predict priority, which was also evaluated using MAE and MSE. The benchmark was performed on 2,225 situations, corresponding to the full dataset. Each model produced one prediction per situation. Although this evaluation configuration does not take stochastic variability into account, it still allows meaningful relative comparisons between models to be made.

E. Agent Architecture with LangGraph

The different components of SAGA are integrated into an agent pipeline using LangGraph. The agent maintains an explicit internal state, which includes the situation description, extracted PCS, predicted priority, decision outcome, and interaction history. The pipeline is structured into four main nodes:

- Perception : which extracts PCS from the textual description using an LLM.
- Reasoning : which predicts priority from PCS using the Random Forest model and derives a simple decision.
- Explanation : which generates a human-readable justification based on dominant PCS dimensions.
- Negotiation : which is activated only if the user disagrees with the proposed decision.

This modular structure makes the agent behaviour transparent and easy to analyse, while also allowing future extensions.

F. User Interaction and Negotiation

A simple user interface was developed using Streamlit to interact with the SAGA agent. The interface allows users to describe two situations they are considering, select the LLM they wish to use, and observe the agent's reasoning process. Each situation is analysed independently, and the agent recommends the meeting with the highest predicted priority. If the user accepts the recommendation, the interaction ends. If the user refuses, the agent enters a negotiation phase. In this phase, an LLM generates a short counter-argument that acknowledges the user's concern and insert the most relevant psychological characteristics of the situation in its argument. The negotiation history is stored to avoid repeating arguments.

IV. Experimental Setup and Results

This section presents the results obtained with the proposed SAGA architecture.

A. Evaluation Setup

All models were evaluated on the same dataset composed of 2,225 situations. For each situation, a textual description was provided as input to the LLM, which produced PCS scores for the eight DIAMONDS dimensions. These predicted scores were then compared to the ground truth annotations from the dataset. To evaluate performance, Mean Absolute Error (MAE) and Mean Squared Error (MSE) were computed for each PCS dimension. In addition, the predicted PCS values were passed to the Random Forest Regressor to estimate the situation priority. The resulting priority predictions were compared to the reference priority values using the same error metrics.Each model was evaluated using identical prompts and parameters. No fine-tuning was applied. The entire benchmarking system was created in a Jupyter Notebook, while the SAGA agent was coded in Python and divided into several modules. To install Jupyter and the other libraries used in the project, run the following command:

```
pip install jupyter
```

I also used the following libraries : numpy, pandas, tqdm, langchain_groq, google, sklearn, joblib, ollama, streamlit and langgraph. LangGraph allows you to create agent pipelines. Ollama allows you to install LLMs locally on your machine. I also used the Groq and Google APIs to evaluate LLMs online. You also need to install Ollama from the official website and add a new environment variable containing the location of the models you are going to install. This folder can be specified in the Ollama settings. To use

Groq, simply create your API key and specify your model when calling the API with the langchain_groq library. Follow the same instructions to use the Google API. Furthermore, to install a model locally with Ollama, run the following command:

```
ollama pull [insert model name]
```

Finally, to launch the SAGA agent, you must run this command :

```
streamlit run app.py
```

B. Benchmark Results

Table I reports the PCS extraction results for all evaluated models. For each DIAMONDS dimension, MAE and MSE are reported. Lower values indicate better performance, with MAE reflecting average deviation from the ground truth and MSE penalizing larger errors more strongly.

The benchmark conducted in this project remains limited due to several practical constraints. Initial experiments attempted to include large online models such as LLaMA 3.3 70B using the Groq API. However, the daily request limits were quickly reached, making it impossible to process the full dataset of 2,225 situations. Similar issues were encountered when using the Google Gemini API. In addition, limited local hardware resources and a poor network connection restricted the number of models that could be installed and evaluated locally. As a result, the benchmark focuses mainly on small to mid-sized models that were practically usable in this context. In addition, I also tried using Phi 3.5, but the outputs were inconsistent and returned a lot of unwanted information or formatting errors, which generated numerous errors and rendered the results unusable.

Overall, the results show consistent and interpretable differences between models. No single model consistently dominates all PCS dimensions, which suggests that different psychological characteristics pose different levels of difficulty for LLM-based extraction. Among the evaluated models, Gemma 2 2B and Gemma 2 9B exhibit the most balanced performance across PCS dimensions. While Gemma 2 9B shows slightly better stability on socially nuanced dimensions such as Deception and Sociality, Gemma 2 2B remains competitive despite its smaller size. Llama 3.2 3B shows moderate performance overall, with higher errors on dimensions such as Mating and Sociality, suggesting some difficulty in capturing relational or interpersonal cues from the descriptions. Mistral 7B generally performs worse than the other models across most dimensions, particularly on Negativity and Duty, where both MAE and MSE are noticeably higher.

Some PCS dimensions appear consistently easier to predict across models. Intellect and Adversity show relatively low MAE values for all models. This suggests that these dimensions are grounded in explicit cues present in the descriptions. Mating remains one of the most difficult dimensions, especially for smaller models such as Llama 3.2 3B, which reaches an MAE above 2. This confirms that romantic or attraction-related cues are harder to infer, particularly when they are weakly expressed in text. Deception and Sociality display moderate variance across models. Gemma 2 9B achieves the lowest MAE on Deception (0.91), while other models show higher but still reasonable errors.

Across all PCS dimensions, the lowest average MAE is obtained by Gemma 2 9B, closely followed by Gemma 2 2B. This result suggests that PCS extraction does not strictly require large-scale models. In contrast to PCS extraction, priority prediction errors are relatively similar across models. All evaluated models achieve a Priority MAE between approximately 1.5 and 1.8, with MSE values clustered around 4.4. The best-performing model in terms of priority MAE is Gemma 2 2B (MAE = 1.54), closely followed by Llama 3.2 3B and Gemma 2 9B.

This result highlights an important observation: differences in PCS extraction accuracy do not translate linearly into priority prediction performance. Even models with moderate PCS errors can lead to comparable priority predictions. This behaviour is likely explained by the robustness of the Random Forest regressor, which aggregates multiple PCS dimensions and can tolerate noise in individual features.

In the reference paper, priority prediction based on ground-truth psychological characteristics of situations achieves an MAE of 0.98 on a 7-point Likert scale. When psychological characteristics are themselves predicted from social situation features, the MAE increases to 1.37, which remains comparable to the baseline model relying directly on social situation features (MAE = 1.35). In our

TABLE I
BENCHMARK RESULTS OF SAGA

Dimension	Llama 3.2 3B		Gemma 2 9B		Gemma 2 2B		Mistral 7B	
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
Duty	1.98	6.3	2.17	6.36	1.92	5.71	2.7	10.66
Intellect	1.74	4.7	1.79	4.52	1.51	3.57	1.34	2.78
Adversity	1.4	3.33	1.39	4.14	1.43	4.19	2.19	6.93
Mating	2.12	8.24	1.8	4.39	1.25	3.52	**0.93**	2.48
Positivity	1.92	5.95	1.45	4.13	1.62	4.51	2.33	7.12
Negativity	1.38	3.83	1.32	3.5	1.52	4.52	2.98	11.26
Deception	1.33	3.6	**0.91**	2.73	**1.08**	3.23	2.14	5.63
Sociality	2.2	7.25	1.33	3.41	2.19	2.19	1.74	5.11
Priority	**1.56**	**4.43**	**1.59**	**4.4**	**1.54**	**4.38**	**1.76**	**4.44**

experiments, priority prediction relies on PCS values extracted from natural language descriptions rather than on human-annotated PCS. As shown in Table I, the resulting MAE values range between 1.54 and 1.76 depending on the LLM used, with the best performance obtained by Gemma 2 2B (MAE = 1.54) and Llama 3.2 3B (MAE = 1.56). These results are slightly worse than those reported for predicted PCS in the original study, but remain within the same order of magnitude.

Beyond the numerical results, the behaviour of the SAGA agent was analysed through interactive use. In most cases, the agent produces consistent explanations that reflect the dominant PCS dimensions fairly well. The explanation module generally highlights a high level of duty or adversity when the priority is high, which corresponds to the decision-making logic of the Random Forest model, as already demonstrated in the paper. During negotiation, the agent adapts its arguments to the user's comments. Since the local models used are small models, the arguments often remain generic, especially when the user's objection is vague and mentions names unknown to the model.

V. LIMITATIONS AND FUTURE WORK

While the proposed SAGA framework shows promising results, several limitations must be acknowledged.

A. Generated Situation Descriptions

One important limitation of this work lies in the use of synthetic situation descriptions. The textual inputs were generated from structured data using a large language model, rather than collected from real users. Although this approach ensures consistency with the original dataset, it can reduce linguistic diversity, which is unique to humans, and realism. Future work should evaluate SAGA on descriptions written by humans or in studies conducted with real users to better assess its robustness. However, real studies involving humans are much more difficult because they require much more time and financial resources.

B. Dependence on Prompt Design and Model Training

The PCS extraction relies on prompt engineering. The prompt explicitly defines each DIAMONDS dimension and constrains the output format. While this improves stability, it also introduces a dependency on prompt quality. Small changes in wording or structure may influence the extracted scores, especially for smaller models. Future work could explore prompt sensitivity analysis or compare prompt-based extraction with classical NLP approaches, such as supervised regression models trained directly on text. Furthermore, the training data of LLMs and their hyperparameters greatly influence the results.

C. Limited Evaluation of Stochastic Variability

Each situation in the benchmark was processed once per model. This evaluation configuration does not capture the stochastic variability of LLM outputs. A more rigorous evaluation would involve multiple runs per situation and statistical analysis of variance. However, this was not feasible given the time and material resource constraints. Future experiments could investigate the stability of PCS extraction across repeated generations.

D. Fine-Tuning Models

The priority prediction relies on a pre-trained Random Forest model from the reference paper. While this ensures comparability, it also limits adaptability. The model is not updated or fine-tuned based on the extracted PCS distributions produced by LLMs. The results of LLMs also depend on the data on which they were trained. Certain LLMs could also be fine-tuned to achieve better PCS extraction.

E. Agent Interaction and Negotiation Limitations

The negotiation module is intentionally simple. Arguments are generated based on dominant PCS dimensions and stored in a short history to avoid repetition. However, the agent does not model long-term user preferences or learning across interactions. More advanced negotiation strategies could incorporate user profiles, reinforcement learning or explicit preference modelling. In addition, evaluation of negotiation quality remains qualitative and subjective, which limits the strength of arguments.

F. Future Research Directions

Several extensions can follow this work like evaluating SAGA in personal assistant scenarios with human users or comparing LLM-based PCS extraction with traditional NLP methods and techniques trained directly on text.

VI. Conclusion

In this paper, I presented SAGA, a Situation Aware Generative Agent designed to support decision-making in social scheduling contexts. The main objective was to explore whether large language models can be used to extract Psychological Characteristics of Situations from natural language descriptions, and whether these extracted features can be integrated into a classical machine learning pipeline for priority prediction. By grounding the agent in the DIAMONDS taxonomy and relying on an existing Random Forest priority model, this work remains closely aligned with prior research on social situation awareness. At the same time, it extends this line of work by introducing LLMs as a perception layer, capable of translating unstructured descriptions into structured psychological representations. The benchmarking experiments show that different LLMs show varying levels of accuracy and stability when performing PCS extraction. Smaller models can achieve interesting results under constrained settings. These findings suggest that LLM-based perception is a viable approach, even when computational resources are limited. Beyond the quantitative evaluation, this project also demonstrates how PCS-based reasoning can be embedded into an interactive agent architecture. Using LangGraph and a lightweight negotiation mechanism, SAGA can recommend a meeting, explain and defend its decision in a human-understandable way. Finally, on a personal level, this project was a valuable learning experience. It allowed me to work with tools such as LangGraph, Streamlit, local LLMs, and external APIs, while also strengthening my Python skills. These technical skills will be useful for my future career as an engineer.

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